Аналитика и модель

Индекс → признаки → target → baseline → ограничения

Что нужно знать

- Формула Champagne Index; почему goal + assist может учитывать одно взятие ворот дважды и не является xG.
- Target = индекс следующего последовательного сезона минус текущий; минимум 30 матчей для топ-5 лиг и 5 для прочих соревнований.
- Пять признаков: текущий индекс, общие расходы, расходы на атаку, средний возраст, изменение средней рыночной стоимости.
- Как устраняется неверный лаг: уникальный клуб-сезон, точный предыдущий год, затем присоединение соревнований.
- Хронологический holdout: ранние сезоны обучают, поздние тестируют; baseline предсказывает нулевое изменение.
- R^2 = 0.127 не означает 87% или 12.7% точности. MAE 0.4856 против 0.5114 показывает небольшое улучшение.
- Lab использует отдельную модель на всех допустимых данных и не доказывает причинный эффект изменения бюджета.
- The season comparison also shows score-derived goals for/against/difference and per-match attack/defense ranks. These are separate from the Transfermarkt-based Champagne Index.
- The defensive-spend follow-up correlates defender fees with next-season change in goalkeeper-source goals conceded per match: r=0.0040, p=0.427, n=38,531; no useful linear signal.

Что показать

1. Показать текущие r, p, n и MAE на странице Project; назвать единицы и выборку каждого показателя.
2. Объяснить, почему 23/24 → 25/26 не является корректной парой, даже если shift(-1) её возвращает.
3. Поменять один ввод в Lab, вернуть baseline и объяснить, что измеряется сравнением сценариев.

Короткая речь на английском

> *We define attacking output as recorded goals plus assists per source match. The target is the change in that index in the next consecutive season. We remove incomplete season pairs and construct lagged squad values before joining competitions. Evaluation uses earlier seasons for training and later seasons for testing. The RandomForest has a test MAE of about 0.486, compared with 0.511 for predicting no change: an improvement of roughly five percent. This is a modest signal. The comparison also separates the Transfermarkt attacking index from completed league scores: goals for, goals against, their difference, and per-match attack and defense ranks. A defensive-spend follow-up also has almost zero association with next-season conceded-goal change (r=0.004, p=0.427; n=38,531). The Lab uses a separate all-data model for retrospective exploration. Moving a budget slider changes a model input; it does not establish what a transfer would cause.*

Переход: “The application makes those distinctions visible and lets us inspect the evidence for individual clubs.”

Практические задания

1. Рассчитай (76 + 70) / 38 и объясни, почему ответ 3.84 не равен голам за матч.
2. Посчитай (0.511364 − 0.485593) / 0.511364 ≈ 5.04% и назови правильную интерпретацию.
3. Различи три вопроса: корреляция в текущем сезоне, изменение в следующем сезоне и причинный эффект покупки игрока.

Вопросы по направлению

Is Champagne Index an official metric or xG?

Это официальный индекс или xG?

No. It is this project’s proxy: recorded goals plus assists divided by source matches. A goal with an assist contributes twice. We explain the denominator and do not present it as xG or an official rating.

What exactly does the model predict?

Что именно предсказывает модель?

The next consecutive season’s Champagne Index minus the current index. It does not predict a match score, a calibrated league position or the causal return from buying a player.

Why reject gaps between seasons?

Почему нельзя брать просто следующую строку?

A club may leave a competition and return years later. shift(-1) gives the next recorded row, which is not necessarily next year. We explicitly require a one-year gap and adequate coverage in both seasons.

Why use a temporal split?

Зачем хронологический holdout?

A random row split mixes earlier and later seasons and can make historical evaluation overly optimistic. We train on earlier seasons and test on later ones, keeping observations from the same season together.

How much better is the model than a baseline?

Насколько модель лучше baseline?

On the same 5,585 held-out rows, MAE is about 0.4856 versus 0.5114 for predicting no change. That is about a five percent reduction in MAE, a modest gain. R-squared is about 0.127, not a percentage of correct predictions.

Can you conclude that buying attackers improves performance?

Можно ли утверждать причинный эффект?

No. These are retrospective observational data with confounding and source limitations. A slider changes a model input and shows a learned association, not the outcome of a controlled transfer intervention.

Why is the Lab model different from the evaluation model?

Почему две обученные модели?

The evaluation model uses only earlier training data and is scored on held-out later seasons. A separate model is refitted on all eligible data for exploration. Its own training fit is not claimed as independent test performance.

Why did the index-position correlation change after the audit?

Почему изменилась корреляция?

The earlier result mixed a partial 25/26 attacking snapshot with full final standings. We now require adequate comparable coverage. The corrected same-season association is about minus 0.793 over 388 observations.

What is the main result in one sentence?

Главный результат одним предложением?

The system is reproducible on the measured setup, while historical transfer spending alone has almost no linear relationship with next-season index change and the richer model adds only modest predictive value.

What is the most important limitation?

Главное ограничение?

The data are retrospective and unevenly covered, the index is a proxy, and associations are not causal effects. Source licence and independent team/laptop validation also remain explicit unresolved requirements.

How do goals for, goals against and goal difference work in the comparison?

Как в сравнении рассчитываются забитые, пропущенные голы и разница?

GF and GA are season totals rebuilt from completed home/away match scores. GD is GF − GA. They come from the optional football-data.co.uk match enrichment, not from the Champagne Index player totals.

How are attacking and defensive ranks calculated?

Как рассчитываются атакующий и защитный рейтинги?

Within one league and season, attack rank orders goals scored per match from highest to lowest; defense rank orders goals conceded per match from lowest to highest. SQL RANK gives tied rates the same rank and leaves a gap after ties. They describe scores, not tactics or chance quality.

Do we have xG or can we estimate it?

Есть ли в данных xG или можем ли мы его оценить?

The source has no provider xG field. The comparison instead shows observed shots and shots on target. A separate estimate multiplies shots on target by the league-season average goals-per-shot-on-target rate; it is an experimental volume proxy with no shot-location or chance-quality input, not xG.

What is the difference between the Champagne Index and attack rank?

Чем атакующий рейтинг отличается от Champagne Index?

The Champagne Index is recorded player goals plus assists divided by a Transfermarkt matches proxy for a club-season. Attack rank uses actual completed league scorelines, sums team goals and ranks goals per match within the league-season. Their sources, coverage and units differ.

How did you check the goalkeeper defensive totals?

Как вы проверили вратарские показатели по пропущенным?

We compared Transfermarkt goalkeeper goals-conceded totals with the independent match-result totals on 388 adequately covered top-five club-seasons. They match exactly in 366 (94.3%). Clean sheets match only 269 (69.3%), so the scoreline source remains the basis of league ranks.

What does the defensive-spend analysis show?

Что показывает анализ расходов на защитников?

The Pearson correlation between defender transfer fees and next-season change in the goalkeeper-source conceded-goal rate is r=0.004047, p=0.426997, n=38,531. The sample does not support a useful linear association. It is descriptive, not causal. A separate match-level check finds teams with a red card concede 0.341 more goals on average in the same match, but match context confounds that association.

Why keep the player-source defensive values separate from scoreboard goals?

Почему defensive-показатели по игрокам отделены от итогов матчей?

The sources use different records and denominators. Goalkeeper conceded totals agree with scorelines in 94.3% of adequately covered seasons, while clean sheets agree in 69.3%. The UI therefore uses match scores for GF, GA, GD and ranks; player-source totals remain a quality cross-check. Shots and corners are also observed match statistics, not chance-quality measures.

Is the shot-based estimate the same as xG?

Оценка по ударам — это xG?

No. It multiplies a team’s shots on target by the league-season average goals per shot on target. It has no shot-location, distance, body-part, pressure or goalkeeper-position input. It is a rough volume benchmark and must be called an SOT-based proxy, not provider xG.

What does the Poisson attack/defense rank tell us?

Что показывают Poisson-ранги атаки и обороны?

The model fits attack, defense and a shared home-advantage term to the selected league-season scores. Larger fitted attack and defense parameters mean stronger estimated attack and defense. Its ranks are descriptive and in-sample; the separate chronological RandomForest evaluation is the project’s held-out prediction check.

Did the added defensive features improve the model?

Улучшили ли защитные признаки модель?

On the same chronological split, R^2 moves from 0.1271 with the five-feature baseline to 0.1502 after adding goalkeeper conceded rate, discipline and net spend; MAE moves from 0.4856 to 0.4788 on 5,585 held-out rows. The gain is modest, and it does not isolate a causal effect of defense.

Do red cards cause more goals to be conceded?

Красные карточки приводят к большему числу пропущенных?

The local sample shows a same-match difference of +0.341 conceded goals per team-match (1,619 sent-off vs 16,195 others; Welch $p < 0.0001$). This is observational: stronger opponents, match state and timing can affect both red cards and goals. It is not a causal estimate.

Какие фразы не использовать

- Модель доказывает, что увеличение бюджета улучшит игру.
- R^2 — доля правильных прогнозов.
- Оценка отдельной holdout-модели является результатом тестирования all-data модели Lab.

> Роль — учебная ответственность. Не заявляй личное авторство или самостоятельную проверку, если этого не было. Покажи ровно те изменения и команды, которые выполнял(а) лично.